Reading Both Halves of *Imvo*: What Modernising 1880s isiXhosa Does to Cross-Lingual Analysis

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Abstract

***Imvo Zabantsundu* (1884–1894) printed a large isiXhosa section and a smaller English one that was not a full translation of it. This study tests whether a multilingual encoder (LaBSE) can see this difference once language is removed, and whether modernising the 1880s spelling changes the answer. The newspaper's own translations are the control: a classifier's accuracy on them, the floor, counts as language, not content. Above that floor the sections clearly differ, in every setting tried. Modernising has an inconclusive effect on this gap and moves the floor, not accuracy on all text: spelling changes how well the tool reads the paper, not what it says. It also rewrites the names of peoples, places and God.**

Keywords: historical text normalisation; multilingual sentence embeddings; isiXhosa

Introduction

Imvo Zabantsundu, founded by John Tengo Jabavu in 1884, was the first African-owned and run newspaper in South Africa. It was printed mainly in isiXhosa, with an English translation of the editorial and, at times, other English articles, letters and adverts (Schoots, Ngwendu, et al., 2023). A bilingual reader can see that the sections differ; this study asks whether NLP tools, built mostly from modern, high-resource text (Joshi, Santy, et al., 2020), can see it too, because that decides whether archives like this can be studied at scale. The tools also invite researchers to "clean" the isiXhosa first by modernising its spelling: the archive writes *Tixo*, *apa* and *umteto* where modern isiXhosa writes *Thixo*, *apha* and *umthetho*.

Research question. *If the part of a multilingual encoder's embeddings that only encodes language is removed, can a classifier still tell Imvo's isiXhosa and English sections apart by their content, and does modernising the 1880s isiXhosa spelling change how large that content difference looks?* **H1:** a classifier tells the sections apart more accurately on ordinary chunks (60-word pieces of a page) than on translated pairs, which say the same thing; this gap, the measure of content difference, is expected to be above zero. **H2:** modernising lowers the accuracy on translated pairs (the floor), because the encoder reads modern spelling better, but leaves the gap unchanged, within ± 0.02 (measurement invariance); otherwise, a cleaning step changes the conclusion.

Background The archive was OCR-corrected by first-language isiXhosa speakers and deliberately not spell-checked, because 1880s spelling differs from today's (Schoots, Ngwendu, et al., 2023). Historical normalisation is usually judged by the accuracy of its spellings or by task scores (Bollmann, 2019), rarely by whether it changes a conclusion, which makes it a "forking path" (Gelman & Loken, 2013). LaBSE (Feng, Yang, et al., 2022) was trained to match translations in 109 languages, isiXhosa among them, but still encodes language, which can be removed by subtracting each language's average embedding (Libovický, Rosa, et al., 2020) or by projecting out a few directions (Yang, Yang, et al., 2021; Ravfogel, Elazar, et al., 2020). Subtracting averages also removes any average content difference, so a linear classifier would find none. Following probing work that asks for a control (Hewitt & Liang, 2019), this study uses the newspaper's own translations.

Methods

Data. The 511 of 517 Imvo issues in the available copy with at least 1,000 words per language are used: 3.9M isiXhosa and 1.5M English words, close to the archive's counts (Schoots, Ngwendu, et al., 2023). OCR artefacts but never spelling are fixed, pages are cut into 60-word chunks, and boilerplate (text repeated across many issues, such as standing adverts: 29.9% of isiXhosa and 7.9% of English chunks) and chunks in the wrong language are dropped, leaving 45,465 isiXhosa and 22,927 English chunks. Issues are split 357/77/77 into train/validation/test, stratified by year.

Modernisation. From 3.3M tokens of modern isiXhosa (88.5% OPUS-100 (Zhang, Williams, et al., 2020), the rest Wikipedia and Vukuzenzele (Lastrucci, Rajab, et al., 2023)) a word list is built. *Aspiration* adds the missing *h* after *t, p, k, r, c, q, x*, but only if the old form is almost absent from the list ($\leq$2 uses) and the new one is present ($\geq$5). *Prefix* joins prefixes written apart (*e Palamente*$\rightarrow$*ePalamente*) only if the joined form is in the list ($\geq$5 uses). These thresholds were fixed in advance to favour precision: changing a correct word is worse than missing a change. The rules rewrite 9.1% of the 3.7M isiXhosa words (7.8% by aspiration), including 191 words the authors capitalised in $\geq$90% of non-headline uses (19,917 rewrites): names such as *Xosa*, *Tembu*, *Mtata* and *Tixo*, and institutions such as *Rulumente*.

Translated pairs. Chunks are embedded with LaBSE and, as a check, multilingual-E5 (Wang et al., 2024). Two chunks from the same issue, one per section, that are each other's nearest neighbour form a candidate translation; LaBSE candidates are kept with similarity $\geq$0.55 (and repeat at $\geq$0.65). Pairs were not hand-labelled, so this threshold is provisional, but names and numbers, which are not translated, support it: kept pairs share three or more nine times as often as dropped ones (25.7% vs 2.9%). E5's similarities are squeezed into 0.83–0.88, so for E5 the same share of candidates is kept.

Measuring content. A translated pair says the same thing twice, so the difference between its embeddings is mostly language. These differences are stacked for training pairs and their top k directions are removed from every embedding:

$$D = \left[x_i^{\text{xh}} - x_i^{\text{en}}\right]_{i \in P}, \qquad \tilde{x} = x - V_k^{\top} V_k x, \tag{1}$$

where V_k holds the top-k right singular vectors of D. A class-balanced logistic regression then predicts each chunk's section, scored by balanced accuracy (BA; chance is 0.5). Its BA on held-out translated pairs is the *floor*, and the *gap* is $\Delta = \text{BA}_{\text{all}} - \text{BA}_{\text{pairs}}$. On validation, $k \in \{0, 1, 2, 4, \ldots, 64\}$ is chosen as the smallest with a floor $\leq$0.55 (else the largest), as is the regularisation strength $C \in \{10^{-2}, \ldots, 10^{3}\}$. A cosine k-nearest-neighbour (kNN) probe repeats the measurement as a non-linear check. For H2, variants reuse the raw

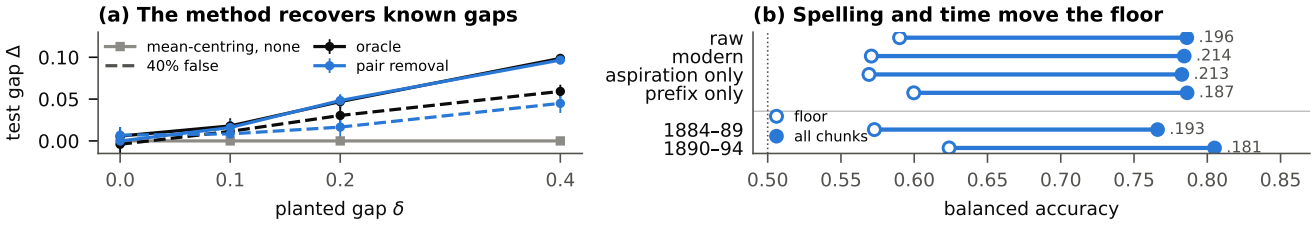


Figure 1: (a) Synthetic data (5 seeds, s.d.; dashed: 40% false pairs). (b) Test issues, LaBSE: modernising shifts the floor, not all-chunk accuracy; across the decade both rise while Δ holds.

Table 1: Section probe on test issues, LaBSE (balanced accuracy).

Variant	All	Floor	Δ	$\Delta - \Delta_{\text{raw}}$ (90% CI)
raw	.786	.590	.196	–
modern	.784	.571	.214	$+$.018 ($-$.003, $+$.037)
aspiration only	.783	.569	.213	$+$.017 ($-$.001, $+$.038)
prefix only	.786	.600	.187	$-$.009 ($-$.024, $+$.006)

pairs and k, and a variant is equivalent to raw if the 90% CI of its change in Δ (1,000 bootstraps over test issues) lies within $\pm$0.02 (two one-sided tests, α=0.05). Topics are k-means clusters of $\tilde{x}$, fitted without section labels. **Synthetic check.** To test the method on a known answer, embeddings are simulated as a content (topic) vector plus a language vector. English chunks draw topics evenly; a share δ of isiXhosa chunks draws from only a quarter of them, a content gap that is *planted*. Translated pairs copy isiXhosa content into English. An *oracle* probe sees only the content, giving the true gap.

Results

The method finds planted gaps. On synthetic data (Figure 1a), pair removal recovers the planted gap (0.097 against the oracle's 0.099 at δ=0.4) and, with 40% false pairs, underestimates it (0.045 vs 0.059) rather than inflating it; no removal and mean-centring report 0.

The sections differ (H1). Removing one language direction drops the validation floor from 0.97 to 0.59, but no k brings it to 0.55, so k=64 is used (test floor 0.590, 95% CI 0.561–0.621). Above this floor the gap is 0.196 (0.167–0.224), or 0.258 with E5. It holds with stricter pairs (0.163), a kNN probe (0.149; 0.046 with E5) and without number-heavy chunks such as price lists (0.185, 22% of isiXhosa chunks removed), and it is positive in all 56 settings run; only mean-centring with the linear probe removes it. It is also a lower bound: the floor is 0.64 on the least similar half of test pairs but 0.54 on the most similar, giving a gap of 0.244 (Figure S1).

Spelling moves the floor (H2). No spelling variant is equivalent to raw or clearly different from it (Table 1; with E5 the change is $+$0.013, $-$0.015 to 0.039): every 90% CI crosses $\pm$0.02 but includes 0. Accuracy on all chunks does not change ($-$0.002, 95% CI $-$0.007 to 0.004); the change comes from the floor ($-$0.019; Figure 1b), and translated pairs move closer together (similarity $+$0.005, 0.003–0.007) although they were chosen on raw text. Time behaves the same way for LaBSE: all-chunk accuracy and the floor both rise over the decade (Figure 1b), and against each period's own floor Δ changes by $-$0.012 (CI $-$0.073 to 0.048); for E5 it is inconclusive. H2 is neither supported nor rejected.

Discussion

An encoder can see that Imvo's readerships got different content, and the newspaper's own translations show it by separating how well the tool reads the paper from what the paper says. The changes tested moved the first, not the second. Modern spelling turns words absent from modern text into ones that occur in it (31.5% of words absent before aspiration, 24.5% after), so an isiXhosa chunk lands closer to its English translation and less language is left for the probe, while all-chunk accuracy stays put because the content did not change. The floor also explains why no k reaches 0.55: what remains is mostly loose pairs, which differ in content too, so subtracting the floor makes the gap conservative.

The encoder sees this content without supervision. LaBSE was trained to match translations, not to sort topics, yet k-means on its language-removed vectors separates each issue's halves twice as much as shuffled sections (Jensen–Shannon divergence between their topic mixes, 0.141 vs 0.067). Each readership got content the other rarely saw: 16% of isiXhosa chunks fall in topics holding 1.9% of English chunks, such as market prices (*Inkuku 7d. ne 8d. inye*, "chickens 7d. and 8d. each"), land notices and sport, and 32% of English chunks in English-majority topics such as liquor licensing (Figure S2; silhouette 0.008: overlapping clusters). The English section was a second paper, not just a translation.

Modernising has a cost the gap does not show: it rewrites the names of a people (*Xosa*), a community (*Tembu*), a town (*Mtata*) and God (*Tixo*). "Modern" is also set by available data (here 88.5% OPUS-100), and tools that favour English (2.11 LaBSE tokens per isiXhosa word, 1.26 per English one) tempt researchers to study only the English half. The conclusions are limited by unchecked pairs and rewrites, a floor above chance and pairs mostly from page 2 (72% vs 49% overall); the method measures content, not intent.

Conclusion

An encoder can see that Imvo's readerships got different content, but only against the floor its own translations give; archive studies should report this floor on raw and normalised text.

Contribution Statements

We, Gopolang Mohlakola, Tawananyasha Kadango and Lily de Melo, worked together fairly on this project.

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    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: The code, normaliser and labelling guide are documented in README.txt and labels/LABELLING_GUIDE.md.

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    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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    Answer: [NA]

    Justification: No human subjects: the texts are newspapers from 1884–1894, and the archive's authors note that no ethics review was needed because the materials discuss no living subjects (Schoots et al., 2023).

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Wits Faculty of Science AI Ethics Statement

Wits University Faculty of Science student AI declaration. We understand that the use of generative AI tools (such as ChatGPT or similar) without explicitly declaring such use constitutes a form of plagiarism and is classified by Wits University as academic misconduct. We declare that in the course of conducting this research and preparing this report:

[ ] We did not make use of generative AI tools.

[X] We did make use of generative AI tools for the following:

- [X] 1. Idea generation (research problem/design, hypothesis)
- [X] 2. Sourcing related work (summarising, identifying sources)
- [X] 3. Methods and experiment design (experiment setup, model tuning)
- [X] 4. Data analysis (presentation, coding, interpretation)
- [X] 5. Theoretical development (conceptual analysis)
- [X] 6. Code development (generating algorithms, writing scripts)
- [X] 7. Presentation (rendering graphics, formatting)
- [X] 8. Editing (grammar, readability)
- [X] 9. Writing (text generation, document structuring)
- [X] 10. Citation formatting (structuring, organising)

Generative AI tool used: Claude (Anthropic), through Claude Code. **Used for:** refining the research question and hypotheses; designing the controls (translated-pair floor, synthetic check, equivalence test); writing and debugging the pipeline code; running and interpreting analyses; drawing figures; drafting and editing the text; and checking every number and citation against the result files and sources. The AI was used as an assistant throughout the design and execution of the experiments; the authors reviewed all of its output and take full responsibility for all code and writing submitted.

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2545080 (Lily de Melo)	_____	_____

Supplementary material

Two figures behind claims in the body, plotted from the files in `results/`. LaBSE is blue and multilingual-E5 orange.

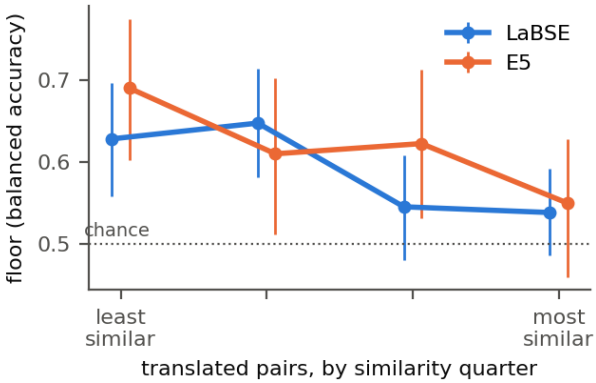


Figure S1: Why Δ is a lower bound. Floor of the section probe by quarter of pair similarity (test pairs, 95% bootstrap CIs over issues). The closer a pair is to a true translation, the closer the classifier is to chance: much of the floor reflects loose pairs, not language.

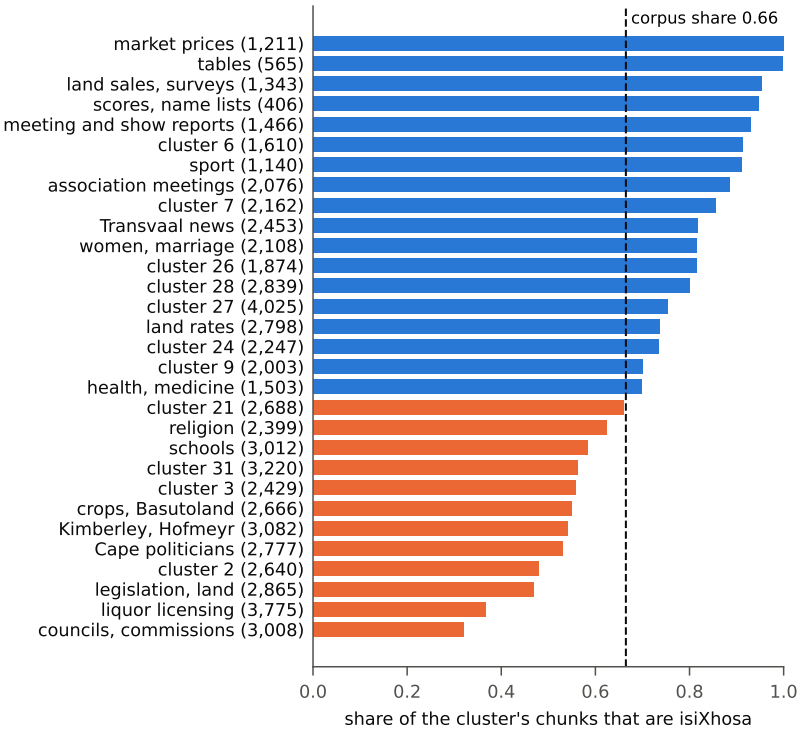


Figure S2: What differs. Share of isiXhosa chunks in each of the 30 topic clusters with more than one chunk (k-means on language-removed LaBSE vectors, fitted on training issues; chunk counts in brackets). Blue: above the corpus share of isiXhosa (0.66); orange: below. Labels are the authors' reading of each cluster's distinctive words; the clusters are weakly separated (silhouette 0.008), so they describe rather than count.